Practice Quiz Nine: Phasors: Phasor Math (Intermediate)

Part One: Given the time-domain expression of a sinusoidal current source below, find the phasor-domain representation I_s in both polar and rectangular form.

$$i_s(t) = 7\sqrt{2}\cos(3{,}000\mathrm{t} + 135^\circ)\ \mathrm{A}$$

Finally, plot I_s on the complex plane.

Part Two: Given the two voltage phasors below, perform the following operations:

a) $V_1 + V_2$

b) $V_1 - V_2$

c) $V_1 V_2$

d) $\dfrac{V_1}{V_2}$

$$V_1 = -5 - 2j$$
$$V_2 = 3\angle{-45^\circ}$$

Finally, write the answers in both polar and rectangular form.

Suggested time limit: 10 minutes